

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 21
 - Riemann-Stieltjes integrals
 - Stieltjes integrals

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 - Riemann-Stieltjes integrals
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Integration by parts

The integrand and integrator play dual roles:

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The integrand and integrator play dual roles:

Theorem

Integration by parts Let f be integrable with respect to α on $[a, b]$.

Then α is integrable with respect to f on $[a, b]$ and

$$\int_a^b f d\alpha = f(b)\alpha(b) - f(a)\alpha(a) - \int_a^b \alpha df.$$

Integration by parts

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The starting place of the proof is the telescoping sum

$$f(b)\alpha(b) - f(a)\alpha(a) = \sum_{k=1}^n [f(x_k)\alpha(x_k) - f(x_{k-1})\alpha(x_{k-1})].$$

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Then the left-hand side of the equality in the theorem motivates

$$\begin{aligned} & f(b)\alpha(b) - f(a)\alpha(a) - S(P, \alpha, f, \{t_k\}) \\ &= \sum_{k=1}^n [f(x_k)\alpha(x_k) - f(x_{k-1})\alpha(x_{k-1}) - \alpha(t_k)f(x_k) + \alpha(t_k)f(x_{k-1})]. \end{aligned}$$



Integration by parts

Proof.

If we think about the partition:

$$P' = P \cup \{t_k\} = \{x_0, t_1, x_1, t_2, x_2, \dots, t_n, x_n\}$$

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$$P' = P \cup \{t_k\} = \{x_0, t_1, x_1, t_2, x_2, \dots, t_n, x_n\}$$

and the sample points:

$$t'_1 = x_0, t'_2 = x_1, t'_3 = x_1, t'_4 = x_2, \dots, t'_{2n} = x_n, t'_{2n+1} = x_n$$

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We can rewrite this as

$$f(b)\alpha(b) - f(a)\alpha(a) - S(P, f, \alpha, \{t'_k\}) = S(P, \alpha, f, \{t_k\}).$$



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Now let $\epsilon > 0$ and set $A = f(b)\alpha(b) - f(a)\alpha(a) - \int_a^b f d\alpha$.

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Choose a partition P_ϵ such that for any refinement P of P_ϵ

$$\left| S(P, f, \alpha, \{t_k\}) - \int_a^b f d\alpha \right| < \epsilon$$

for all choices of sample points.

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for all choices of sample points.

Let P be a refinement of P_ϵ and $\{t_k\}$ be a choice of sample points.

Then for P' and $\{t'_k\}$ defined as above, P' is a refinement of P_ϵ , so

$$|S(P, \alpha, f, \{t_k\}) - A| = \left| S(P', f, \alpha, \{t'_k\}) - \int_a^b f d\alpha \right| < \epsilon$$

Uniform continuity

Definition

Let (S, d_S) and (T, d_T) are metric spaces, $A \subseteq S$, and $f : A \rightarrow T$.

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Let (S, d_S) and (T, d_T) are metric spaces, $A \subseteq S$, and $f : A \rightarrow T$.

Then we say f is **uniformly continuous** on A if for all $\epsilon > 0$ there exists $\delta > 0$ such that

$$d_S(x, y) < \delta \Rightarrow d_T(f(x), f(y)) < \epsilon \quad \forall x, y \in A.$$

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- uniform continuity implies continuity
- uniform continuity = same δ works for all c
- continuity implies uniform continuity on *compact sets*

Uniform continuity

Theorem

Let (S, d_S) and (T, d_T) be metric spaces, $A \subseteq S$, and $f : A \rightarrow T$.

If f is continuous on A and A is compact, then f is uniformly continuous on A .

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Proof.

Assume continuity and compactness, and let $\epsilon > 0$.

Then for all $c \in A$, there exists $\delta_c > 0$ with

$$d_S(x, c) < \delta_c \Rightarrow d_T(f(x), f(c)) < \epsilon/2 \quad \forall x, y \in A.$$

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$$d_S(x, c) < \delta_c \Rightarrow d_T(f(x), f(c)) < \epsilon/2 \quad \forall x, y \in A.$$

Since A is compact, the open cover $\{B_S(c, \delta_c/2) : c \in A\}$ of A has a finite subcover $\{B_S(c_1, \delta_{c_1}/2), \dots, B_S(c_r, \delta_{c_r}/2)\}$ □

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Proof.

Take $\delta = \min\{\delta_{c_1}, \delta_{c_2}, \dots, \delta_{c_r}\}/2$.

Suppose $x, y \in A$ with $d_S(x, y) < \delta$.

Then $x \in B_S(c_j, \delta_{c_j}/2)$ for some j , and therefore

$$d_S(c_j, y) \leq d_S(c_j, x) + d_S(x, y) < \delta_{c_j}/2 + \delta < \delta_{c_j}.$$

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This means that $x, y \in B_S(c_j, \delta_{c_j})$, and therefore

$$d_T(f(x), f(y)) \leq d_T(f(x), c_j) + d_T(c_j, f(y)) < \epsilon/2 + \epsilon/2 = \epsilon.$$

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This means that $x, y \in B_S(c_j, \delta_{c_j})$, and therefore

$$d_T(f(x), f(y)) \leq d_T(f(x), c_j) + d_T(c_j, f(y)) < \epsilon/2 + \epsilon/2 = \epsilon.$$

Since $x, y \in A$ and $\epsilon > 0$ was arbitrary, this proves uniform continuity. □

Relation to Riemann integrals

Theorem

Let f be integrable with respect to α on $[a, b]$.

If α is continuous on $[a, b]$ and continuously differentiable on (a, b) , then $f(x)\alpha'(x)$ is integrable with respect to x on $[a, b]$ and

$$\int_a^b f d\alpha = \int_a^b f(x)\alpha'(x) dx.$$

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Set $M = \max\{|\alpha'(x)| : a \leq x \leq b\}$ and let $\epsilon > 0$.

Since $\alpha'(x)$ is continuous on $[a, b]$, it is uniformly continuous on $[a, b]$.

Therefore there exists $\delta > 0$ with

$$|x - y| < \delta \Rightarrow |\alpha'(x) - \alpha'(y)| < \epsilon/2M(b - a), \forall x, y \in [a, b].$$

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Choose $N \in \mathbb{Z}_+$ with $N > (b - a)/\epsilon$ so that $(b - a)/N < \epsilon$

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Choose $N \in \mathbb{Z}_+$ with $N > (b - a)/\epsilon$ so that $(b - a)/N < \epsilon$

Let $P_\epsilon = \{x_0, x_1, \dots, x_n\}$ with $x_j = (b - a)j/N + a$.



Relation to Riemann integrals

Proof.

Then for any partition P finer than P_ϵ , the Mean Value Theorem says we can choose $\{\tau_k\}$ with

$$f(x_k) - f(x_{k-1}) = f'(\tau_k)(x_k - x_{k-1}),$$

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so that

$$S(P, f, \alpha, \{\tau_k\}) = S(P, f\alpha', x, \{\tau_k\})$$

Relation to Riemann integrals

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Then for any partition P finer than P_ϵ , the Mean Value Theorem says we can choose $\{\tau_k\}$ with

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so that

$$S(P, f, \alpha, \{\tau_k\}) = S(P, f\alpha', x, \{\tau_k\})$$

Therefore for any sample points t_k

$$S(P, f\alpha', x, \{t_k\}) - S(P, f, \alpha, \{\tau_k\}) < \epsilon/2.$$



Relation to Riemann integrals

Proof.

We may also choose P'_ϵ such that for any refinement P of P'_ϵ , we have

$$\left| S(P, f, \alpha, \{t_k\}) - \int_a^b f d\alpha \right| < \epsilon/2.$$

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Proof.

We may also choose P'_ϵ such that for any refinement P of P'_ϵ , we have

$$\left| S(P, f, \alpha, \{t_k\}) - \int_a^b f d\alpha \right| < \epsilon/2.$$

Therefore, for $P''_\epsilon = P_\epsilon \cup P'_\epsilon$, and any refinement P of P''_ϵ :

Relation to Riemann integrals

Proof.

We may also choose P'_ϵ such that for any refinement P of P'_ϵ , we have

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Therefore, for $P''_\epsilon = P_\epsilon \cup P'_\epsilon$, and any refinement P of P''_ϵ :

$$\begin{aligned} \left| S(P, f\alpha', x, \{t_k\}) - \int_a^b f d\alpha \right| &\leq |S(P, f\alpha', x, \{t_k\}) - S(P, f, \alpha, \{\tau_k\})| \\ &\quad + \left| S(P, f, \alpha, \{\tau_k\}) - \int_a^b f d\alpha \right| < \epsilon. \end{aligned}$$



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Concrete Example

Let $f(x) = x$ and $\alpha(x) = x$, and consider $\int_0^1 f d\alpha$.

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If we consider the partition

$$P = \{x_0 = 0/n, x_1 = 1/n, \dots, x_n = n/n\},$$

then for left or right sample points $t_k = x_{k-1}$,

$$S(P, f, \alpha, \{x_{k-1}\}) = \sum_{k=1}^n x_{k-1}(x_k - x_{k-1}) = \sum_{k=1}^n \frac{k-1}{n^2} = \frac{(n-1)}{2n}$$

$$S(P, f, \alpha, \{x_k\}) = \sum_{k=1}^n x_k(x_k - x_{k-1}) = \sum_{k=1}^n \frac{k}{n^2} = \frac{(n+1)}{2n}$$

In either case the limit goes to $1/2$.

Concrete Example

Guess: $\int_0^1 f d\alpha = 1/2$.

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$$S(P, f, \alpha, \{t_k\}) = \sum_{k=1}^n f(t_k) \Delta \alpha_k(P) = \sum_{k=1}^n t_k (x_k - x_{k-1}).$$

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$$S(P, f, \alpha, \{t_k\}) = \sum_{k=1}^n f(t_k) \Delta \alpha_k(P) = \sum_{k=1}^n t_k (x_k - x_{k-1}).$$

Since x is monotone increasing:

$$\frac{n-1}{2n} = \sum_{k=1}^n x_{k-1} (x_k - x_{k-1}) \leq S(P, f, \alpha, \{t_k\}) \leq \sum_{k=1}^n x_k (x_k - x_{k-1}) = \frac{n+1}{2n}.$$

What about other partitions, though?

Stieltjes sums

Let f and α be real-valued functions on $[a, b]$, and let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$ and define

$$m_k(f) = \inf\{f(x) : x_{k-1} \leq x \leq x_k\}, \quad M_k(f) = \sup\{f(x) : x_{k-1} \leq x \leq x_k\}.$$

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The **upper Stieltjes sum** of f with respect to α on $[a, b]$ is:

$$\overline{S}(P, f, \alpha) = \sum_{k=1}^n M_k(f) \Delta \alpha_k(P),$$

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The **upper Stieltjes sum** of f with respect to α on $[a, b]$ is:

$$\overline{S}(P, f, \alpha) = \sum_{k=1}^n M_k(f) \Delta \alpha_k(P),$$

and the **lower Stieltjes sum** of f with respect to α on $[a, b]$ is:

$$\underline{S}(P, f, \alpha) = \sum_{k=1}^n m_k(f) \Delta \alpha_k(P)$$

Stieltjes sums

These sums refine predictably!

Lemma

Let f and α be real-valued functions on $[a, b]$, with α monotone increasing.

Stieltjes sums

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Lemma

Let f and α be real-valued functions on $[a, b]$, with α monotone increasing. If P is a partition of $[a, b]$ and P' is a refinement of P , then

$$\underline{S}(P, f, \alpha) \leq \underline{S}(P', f, \alpha) \leq \overline{S}(P', f, \alpha) \leq \overline{S}(P, f, \alpha)$$

Stieltjes sums

Proof.

The inequality $\underline{S}(P', f, \alpha) \leq \overline{S}(P', f, \alpha)$ is clear.

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We will prove $\underline{S}(P, f, \alpha) \leq \underline{S}(P', f, \alpha)$.

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Let $P = \{x_0, \dots, x_n\}$ and $P' = \{x'_0, \dots, x'_N\}$.

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Let $P = \{x_0, \dots, x_n\}$ and $P' = \{x'_0, \dots, x'_N\}$.

For each k , let $\iota(k)$ with $x_{\iota(k)-1} \leq x'_k \leq x_{\iota(k)}$.

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Let $P = \{x_0, \dots, x_n\}$ and $P' = \{x'_0, \dots, x'_N\}$.

For each k , let $\iota(k)$ with $x_{\iota(k)-1} \leq x'_k \leq x_{\iota(k)}$.

$$m_{\iota(k)}(f) = \inf\{f(x) : x_{\iota(k)-1} \leq x \leq x_{\iota(k)}\} \leq \inf\{f(x) : x'_{k-1} \leq x \leq x'_k\} = m'_k(f)$$

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Proof.

The inequality $\underline{S}(P', f, \alpha) \leq \overline{S}(P', f, \alpha)$ is clear.

We will prove $\underline{S}(P, f, \alpha) \leq \underline{S}(P', f, \alpha)$.

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For each k , let $\iota(k)$ with $x_{\iota(k)-1} \leq x'_k \leq x_{\iota(k)}$.

$$m_{\iota(k)}(f) = \inf\{f(x) : x_{\iota(k)-1} \leq x \leq x_{\iota(k)}\} \leq \inf\{f(x) : x'_{k-1} \leq x \leq x'_k\} = m'_k(f)$$

Therefore

$$\begin{aligned} \underline{S}(P', f, \alpha) &= \sum_{k=1}^N m'_k(f) \Delta \alpha_k(P') \\ &\geq \sum_{k=1}^N m_{\iota(k)}(f) \Delta \alpha_k(P') \\ &= \sum_{k=1}^n m_k(f) \Delta \alpha_k(P) = \underline{S}(P, f, \alpha) \end{aligned}$$



Stieltjes integrals

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The **lower Stieltjes integral** of f with respect to α on $[a, b]$ is:

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Riemann's criterion

Definition

Let f and α be real-valued functions on $[a, b]$.

Then f **satisfies Riemann's criterion** with respect to α on $[a, b]$ if for all $\epsilon > 0$ there exists a partition P_ϵ such that

$$0 \leq \overline{S}(P, f, \alpha) - \underline{S}(P, f, \alpha) < \epsilon.$$

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Theorem

Let f and α be real-valued functions on $[a, b]$, with α monotone increasing. Then the following are equivalent:

- (a) f is Riemann integrable with respect to α on $[a, b]$*
- (b) f satisfies Riemann's criterion with respect to α on $[a, b]$*

Riemann's criterion

Definition

Let f and α be real-valued functions on $[a, b]$.
Then f **satisfies Riemann's criterion** with respect to α on $[a, b]$ if for all $\epsilon > 0$ there exists a partition P_ϵ such that

$$0 \leq \overline{S}(P, f, \alpha) - \underline{S}(P, f, \alpha) < \epsilon.$$

Theorem

Let f and α be real-valued functions on $[a, b]$, with α monotone increasing. Then the following are equivalent:

- (a) f is Riemann integrable with respect to α on $[a, b]$*
- (b) f satisfies Riemann's criterion with respect to α on $[a, b]$*
- (c) $\underline{I}(f, \alpha) = \overline{I}(f, \alpha)$*